

Carleton

# STAT220: INTRODUCTION TO DATA SCIENCE

Fall 2026

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| Meetings | MWF 2a<br>9:50-11am MW \| 9:40-10:40 F | CMC 306 |
| Professor | Amanda Luby<br><br>aluby@carleton.edu | CMC 327 |
| Drop-in | Mon 3-4pm<br>Tues 10:30-11:30am<br>Wed 2-3pm (220 priority)<br>Fri 11:30-12:30 (230 priority)<br>Or by appointment! | CMC 307 |
| Website | github.com/stat220-f26 | |
| Texts | *R for Data Science* (2nd Ed)<br>https://r4ds.hadley.nz/<br>Wickham, Çetinkaya-Rundel, Grolemund<br><br>*Modern Data Science with R* (3rd Ed)<br>https://mdsr-book.github.io/mdsr3e/ | |

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|---|---|
| | Baumer, Kaplan, Horton |
| | *Fundamentals of Data Visualization* |
| | https://clauswilke.com/dataviz/ |
| | Wilke |
| Software | Maize RStudio Server <maize.mathcs.carleton.edu> |
| | R (optional) free from <r-project.org> |
| | RStudio (optional) free from <rstudio.com/downloads> |

## COURSE DESCRIPTION

Stat 220 will cover the computational side of statistics that is not typically taught in an intro or methodology focused course like regression modeling. Most of the data you encountered in your first (or second, or third, ..) stats course were contained in small, tidy .csv files with rows denoting your cases and columns containing your variables. Most of the messiness to these data may have been some missing values (NAs). In this course, we'll learn how to extract information from data in its "natural" state, which is often unstructured, messy and complex. To do this, we will learn methods for manipulating and merging data in standard and non-standard formats, data with date, time, or geolocation variables, text processing and regular expressions, and scraping the web for data. To effectively communicate the information contained in these data, we will cover advanced data visualization methods, including methods for creating interactive graphics. We will primarily use the statistical software R in this course, and cover best practices for reproducible analyses and sharing code.

## COURSE OBJECTIVES

After completing this course, you should be able to demonstrate your competency in each of the following areas:

- **Develop** research questions that can be answered by data
- **Acquire** data by importing different file types into R, accessing data through API's, and scraping data from the web
- **Wrangle** common types of data into the form that is needed for analysis
- **Visualize** data to provide insight and uncover relationships and patterns
- **Communicate** your findings to stakeholders in written or oral format
- **Document** your code and **collaborate** across coding projects.

## COURSE COMPONENTS

*MEETINGS*

There will be three course meetings per week (Mondays, Wednesdays, and Fridays). Daily attendance and active participation is expected. Course meetings will combine demonstrations/lecture and in-class group exercises. On most days, I'll ask you to complete a reading or watch a short video before class. You'll also have a "data science journal" that we'll spend time completing at the beginning or end of each class.

*ASSIGNMENTS*

Homework will be assigned once-ish per week, distributed via GitHub. You will submit homework assignments via gradescope. You will use `quarto` for all assignments and submit all necessary work for each assignment on GitHub.

*PORTFOLIO PROJECTS*

Portfolio project require you to integrate several smaller computational tasks and require clear communication of the proposed solution or findings to a broader audience. You will typically work in pairs.

*LAB QUIZZES*

Part of being proficient in data science is being able to do basic data analysis "on the fly", without access to class resources. There will be 3 short (~30 minute) in-class, pencil-and-paper lab quizzes to assess your ability to recall concepts and describe basic data analysis tasks. I recognize that this won't always map closely to other things we do in this class, so you'll also have the opportunity to revise quiz answers with access to R.

*FINAL PROJECT*

The final project is a capstone experience synthesizing everything you've learned over the course of the term. This is an opportunity for you to exercise your creativity and create something meaningful. The final project is wildly open-ended and more details will follow.

*COMMUNICATION*

Assignments and slides will be shared publicly on our course website. Grades will be posted on gradescope or Moodle. Please use our Ed discussion board for any homework or course content questions; email me privately with any personal matters (grade discussions, illness, emergency, etc.). Any time-sensitive announcements will be sent via Ed. It is your responsibility to make sure that your notification settings allow time-sensitive announcements to reach you.

## GRADING POLICIES

Grades are an imperfect measure of learning, especially when we are all coming to class with different backgrounds and experience levels. This course is designed to reward you for consistently participating, staying on top of the course material, and trying your hardest. The grading scheme in this class is designed to reflect how we learn "in the real world": by making mistakes, receiving feedback, and making changes.

*HOW EACH ASSIGNMENT IS GRADED*

Each assignment will include a short rubric with the specifications of how I will evaluate your work. In general, these are the "marks" that you can receive on each course component:

| | How it's evaluated | How it's recorded |
|---|---|---|
| Homework Problems | Completeness, Correctness, and Effort | Successful or Not Successful |
| Lab Quiz Problems | Completeness and Correctness | Successful or Not Successful |
| Portfolio Projects | Completeness, effort, correctness, and communication quality | Excellent, Successful, or Revision |
| Final Project | Completeness, effort, correctness, and communication quality | Excellent, Succesful, or Not Successful |

*EARNING A COURSE GRADE*

Your course grade is assigned using the table below. Each row indicates the minimum percentage of "Successful" results needed to satisfy the requirement of that grade. To earn a grade, complete *all* requirements listed in the row for that grade.

| | Homework Problems | Lab Quiz Problems | Portfolio Projects (3 total) | Final Project |
|---|---|---|---|---|
| A | 85% | 88% | 2 Excellent, 1 Successful | Excellent |
| B | 75% | 78% | 3 Successful | Successful |
| C | 65% | 68% | 2 Successful | Successful |
| D | 55% | 50% | 1 Successful | Successful |

*Example:* Ben finishes the course with 82% of homework problems successfully completed, 85% of lab quiz problems successfully completed, 1/3 portfolio projects marked "excellent", and a "Successful" final project. Ben satisfies everything in the "B" row and earns a "B" in the course.

*PLUS/MINUS GRADES*

"Plus" and "minus" grades will be given if you complete all the requirements for a base letter grade **and** make sufficient progress toward the next grade. Below is an overview:

- If "B" base grade:
  - *and* A in two bins: "B+"
  - *and* A in three bins: "A-"
- If "C" base grade:
  - *and* at least B in two bins: "C+"
  - *and* at least B in three bins: "B-"
- If "D" base grade:
  - *and* at least C in two bins: "D+"
  - *and* at least C in three bins: "C-"
- If "F" base grade:
  - *and* at least D in two bins: "D-"

*Example:* Mira finishes the course with 88% of homework problems successfully completed, 85% of lab quiz problems successfully completed, 3/4 portfolio projects marked "excellent", and a "Successful" final project. Mira satisfies everything in the "B" row, and meets the "A" threshold for two bins (homework and portfolio projects). Mira earns a B+. If Mira instead receives "excellent" marks on the final project, Mira earns an A-.

*ATTENDANCE AND PARTICIPATION POLICY*

I expect daily attendance and active participation. Missing >5 class meetings or ongoing, consistent issues with being on-task will result in a 1/3 grade deduction. I'll use your data science journal as one measure of engagement: consistent, good-faith entries count toward participation, and a pattern of low-effort or empty entries will factor into the 1/3 deduction in the same way absences do.

*IMPORTANT POINTS ABOUT THIS GRADING SYSTEM*

- Different categories of coursework do not "average together": you can't make up for less-than-great work on portfolio projects by doing very well on quizzes, for instance. Each course grade requires *consistent quality across all bins* to earn the grade
- You do not have to do everything. If you want an "A" in the class, for example, you don't have to complete *every* homework problem correctly, only 85% of them.

*WHAT IF I MISS AN IN-PERSON QUIZ?*

If you need to miss a quiz due to a planned conflict, please make arrangements with me at least a week in advance. If you must miss a lab quiz due to an illness or last-minute emergency, you must notify me in advance of the quiz to arrange an alternative time. If you do not notify me in advance, you will earn 0/10 points on the in-class portion, but can still use a token to complete a revision.

*TOKENS*

- You have four tokens to use throughout the term

- Turning in a token provides either (a) a 72-hour extension on a homework assignment, (b) a revision on a portfolio project, or (c) a retake of a quiz
- Tokens may not be used for final project-related assignments
- You can revise the same portfolio project multiple times, if needed, but you must spend a token each time.
- A portfolio project must be completed with a good-faith effort to be eligible for revision. If I deem a submission to be "not assessable" due to a lack of effort, then it cannot be revised.
- All revisions must be submitted by 11:59pm on the last day of class.

## MATERIALS

*TEXTBOOK*

There is no "perfect" data science textbook. We will use excerpts from the following texts:

- R for Data Science 2e
- Modern Data Science with R 3e
- Fundamentals of Data Visualization

These books are all freely available online. If you prefer a hard copy, they are also available for purchase through the publisher.

*SOFTWARE*

The use of the R programming language, with the RStudio interface is an essential component of this course. You have two options for using RStudio:

1. The server version of RStudio on the web at https://maize.mathcs.carleton.edu. The advantage of using the server version is that all of your work will be stored in the cloud, where it is automatically saved and backed up. This means that you can access your work from any computer on campus using a web browser. The downside is that you have to share limited computational resources with each other!
2. A local version of RStudio installed on your machine.The downside to this approach is that your work is only stored locally, but I get around this problem by keeping all of my work on GitHub. You will learn how to use GitHub throughout the course.

Partway through the course, I will require you to download a local version of R and Rstudio. This is because the amount of computing power we will eventually need is more than the server can handle all at once. Both R and RStudio are free and open-source.

## ACADEMIC INTEGRITY

You are expected to follow Carleton's policies regarding academic integrity. I encourage you to *discuss* homework problems with others to figure out tough

problems. You should code and write up your solutions on your own. Lab quizzes must be done by yourself without communicating with others; all work must be your own. You should collaborate with your teammates on projects, and should use external resources for background research and debugging, but all work should be original. The use of textbook solution manuals (physical or online), course materials from other students, or materials from previous versions of this course are not allowed.

Large-language models (e.g. ChatGPT, Gemini, etc.) should only be used for coding or debugging help after you've attempted to code the problem on your own, and you should never type homework problems directly into a prompt. You should never copy and paste any course materials *into* a large-language model, and you should never copy and paste anything *out of* a large-language model into your course materials. Copying, paraphrasing, summarizing, or submitting work generated by anyone but yourself without proper attribution is considered academic dishonesty (this includes output from LLMs).

I also have a few rules in place to protect my intellectual property. You may not record my lectures using tools such as Otter.ai or upload any video or audio recordings to generate transcripts or study notes. You may not upload my course materials (slides, assignment prompts, note sets, etc.) into AI tools or homework help sites (such as chegg).

"AI" tools are new for all of us and it's OK to have questions about what is and isn't appropriate. Please ask if you are unsure of whether or not your actions are complying with the assignment/quiz/project instructions. Always default to acknowledging any help received. Cases of suspected academic dishonesty are handled by the Provost's Office and I am obligated to report any suspected violations of this policy.

| | Collaboration Allowed |
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| Homework Problems | You are allowed and encouraged to collaborate on homework. You may also use outside resources, but your submitted work must be your own and reflect your own understanding (i.e. it can't be copied). Any sources must be properly attributed. If you cannot explain each line of code to me with just your brain, you should not submit it as your own work. |
| Lab Quiz Problems | **No collaboration is allowed at all** (including online forums like StackExchange or Reddit). |

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| Portfolio Projects | You are expected to collaborate with your group members, but cannot rely on external sources other than to help motivate the questions or provide other background information (including online forums like StackExchange or Reddit). You may use any resources from class and package documentation, but getting answers on significant parts of solutions from outside resources is not allowed. |
| Final Project | You are expected to collaborate with your group, but cannot rely on external sources other than to help motivate the questions or provide other background information. Any outside resources should be properly cited. |

## COMMITMENT TO AN INCLUSIVE AND COLLABORATIVE ATMOSPHERE

We all come to class with different backgrounds and experiences, and this diversity makes our class environment richer. I value diversity and inclusion, and am committed to a climate of mutual respect and full participation in and out of the classroom. This class strives to be a learning environment that is usable, equitable, inclusive and welcoming, regardless of race, ethnicity, religion, gender and gender identities, sexual orientation, ability, socioeconomic background, and nationality. If you anticipate or experience any barriers to learning, please discuss your concerns with me.

## RESOURCES

**ACCOMMODATIONS:** Carleton College is committed to providing equitable access to learning opportunities for all students. The Office of Accessibility Resources (Henry House, 107 Union Street) is the campus office that collaborates with students who have disabilities to provide and/or arrange reasonable accommodations. If you have, or think you may have, a disability, please contact OAR@carleton.edu to arrange a confidential discussion regarding equitable access and reasonable accommodations. You are also welcome to contact me privately to discuss your academic needs. However, all disability-related accommodations must be arranged, in advance, through OAR.

**MATH/STATS SKILLS CENTER:** The MSSC (CMC 331) offers drop-in help R/RStudio help sessions run by friendly and knowledgeable peer tutors Sunday-Thursday from 7-10pm.

**TUTORS:** If you find you need more support than office hours and the MSSC can provide, the Academic Support Center offers peer tutoring on the basis of referrals, requests, and availability of tutors. You can request tutoring through a form on their website, or discuss your needs with me and I can submit a referral.

**TITLE IX:** Please be aware that all faculty are "responsible employees", which means that if you tell me about a situation involving sexual harassment, sexual assault, dating violence, domestic violence, or stalking, I must share that information with the Title IX Coordinator. Although I have to make this notification, you will control how your case will be handled, including whether or not you wish to meet with the Title IX coordinator or pursue a formal complaint.

And finally….

**Take care of yourself.** Do your best to maintain a healthy lifestyle this term by wearing a mask when you're sick, eating a vegetable every now and then, exercising, avoiding excessive drug and alcohol use, getting enough sleep, and taking some time to relax. Your physical and mental health is more important than your grade in this course. There are many helpful resources available on campus and an important part of the college experience is learning how to ask for help. For more information, see Student Health and Counseling (SHAC), the Office of Health Promotion, or the Office of the Chaplain. If you are experiencing physical or mental health symptoms as a result of coursework, please speak with me so we can address the problem together.